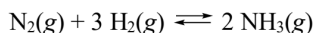


# Equilibrium

1. The reaction of nitrogen and hydrogen to form ammonia is maintained at equilibrium in a rigid container at fixed temperature:



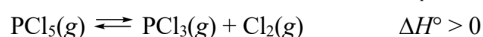
Helium gas is introduced into the system, causing the total pressure to increase by a factor of two. How will the equilibrium for the reaction shift?

- (A) It will be unaffected.  
(B) It will shift to the left.  
(C) It will shift to the right.  
(D) The direction that the equilibrium will shift depends on whether the reaction is endothermic or exothermic.

2. Cacodylic acid,  $(\text{CH}_3)_2\text{AsO}(\text{OH})$ , has a  $\text{p}K_{\text{a}}$  of 6.25. At what concentration of cacodylic acid would a solution of it have a  $\text{pH} = 4.00$ ?

- (A)  $1.0 \times 10^{-4} \text{ M}$  (B)  $3.2 \times 10^{-4} \text{ M}$   
(C)  $5.6 \times 10^{-3} \text{ M}$  (D)  $1.8 \times 10^{-2} \text{ M}$

3. The decomposition of  $\text{PCl}_5$  is endothermic. Which change will result in an increase in the value of  $K_{\text{eq}}$ ?



I. Increasing temperature      II. Increasing volume

- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

4. What concentration of carbonate ion must be achieved to precipitate 99.99% of the silver ion from a 0.100 M solution of silver nitrate? The  $K_{\text{sp}}$  of  $\text{Ag}_2\text{CO}_3$  is  $8.1 \times 10^{-12}$ .

- (A)  $8.1 \times 10^{-10} \text{ M}$  (B)  $8.1 \times 10^{-7} \text{ M}$   
(C)  $1.3 \times 10^{-4} \text{ M}$  (D)  $8.1 \times 10^{-2} \text{ M}$

5. What is the  $\text{pH}$  after 0.100 mol of  $\text{NaOH}$  is added to 250.0 mL of 0.300 M  $\text{H}_3\text{PO}_4$ ? For  $\text{H}_3\text{PO}_4$ ,  $\text{p}K_{\text{a}1} = 2.12$ ,  $\text{p}K_{\text{a}2} = 7.21$ , and  $\text{p}K_{\text{a}3} = 12.38$ .

- (A) 1.82 (B) 2.24  
(C) 6.73 (D) 6.91

6. Zinc hydroxide is a sparingly soluble compound ( $K_{\text{sp}} = 4.5 \times 10^{-17}$ ) that dissolves in strongly basic solution due to the formation of the  $\text{Zn}(\text{OH})_4^{2-}$  complex ion ( $K_{\text{f}} = 5.0 \times 10^{14}$ ). What is the minimum concentration of sodium hydroxide needed in 1.0 L of a solution which could be added to 0.10 mol  $\text{Zn}(\text{OH})_2$  to fully dissolve it?

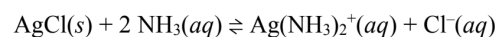
- (A) 2.1 M (B) 2.3 M  
(C) 4.2 M (D) 4.6 M

7. Which statements must be correct at equilibrium?

- I. The rate constants for the forward and reverse reactions are equal.  
II. The rates of the forward and reverse reactions are equal.

- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

8. What is the equilibrium constant expression for the reaction below?



- (A)  $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$  (B)  $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]^2}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$   
(C)  $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{AgCl}][\text{NH}_3]^2}$  (D)  $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{NH}_3]^2}$

9. Phosphoric acid has  $\text{p}K_{\text{a}}$  values of  $\text{p}K_{\text{a}1} = 2.15$ ,  $\text{p}K_{\text{a}2} = 7.20$ , and  $\text{p}K_{\text{a}3} = 12.35$ . Which species is most abundant at  $\text{pH} = 11.5$ ?

- (A)  $\text{H}_3\text{PO}_4$  (B)  $\text{H}_2\text{PO}_4^-$   
(C)  $\text{HPO}_4^{2-}$  (D)  $\text{PO}_4^{3-}$

10. A buffer is made by mixing 0.900 mol  $\text{Na}[\text{CH}_3\text{CO}_2]$  and 0.500 mol  $\text{HCl}$  in enough water to make 2.50 L of solution. What is the  $\text{pH}$  of this solution? The  $\text{p}K_{\text{a}}$  of  $\text{CH}_3\text{COOH}$  is 4.75.

- (A) 4.49 (B) 4.65  
(C) 4.85 (D) 5.01

11. A sealed container with a volume of 1.00 L contains an equilibrium mixture of  $\text{N}_2\text{O}_4(\text{g})$  and  $\text{NO}_2(\text{g})$  at a given temperature. Which quantities will increase if the volume of the container is increased to 2.00 L while maintaining the same temperature?

- I. The number of moles of  $\text{NO}_2(\text{g})$   
II. The partial pressure of  $\text{NO}_2(\text{g})$

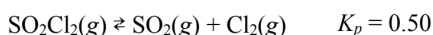
- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

# Equilibrium

12. A solution is 0.05 M in each of the anions fluoride ( $F^-$ ), carbonate ( $CO_3^{2-}$ ), and phosphate ( $PO_4^{3-}$ ). As calcium ion is added slowly to this solution, in what order will the solids precipitate? The  $K_{sp}$  values are:  $CaF_2$ ,  $3.9 \times 10^{-11}$ ;  $CaCO_3$ ,  $3.8 \times 10^{-9}$ ;  $Ca_3(PO_4)_2$ ,  $1.0 \times 10^{-25}$ .
- (A) First  $CaF_2$ , then  $CaCO_3$ , then  $Ca_3(PO_4)_2$   
(B) First  $CaF_2$ , then  $Ca_3(PO_4)_2$ , then  $CaCO_3$   
(C) First  $Ca_3(PO_4)_2$ , then  $CaF_2$ , then  $CaCO_3$   
(D) First  $Ca_3(PO_4)_2$ , then  $CaCO_3$ , then  $CaF_2$
13. The  $K_a$  of formic acid is  $1.8 \times 10^{-4}$ . What is the equilibrium constant for the reaction below?
- $$HCO_2H + OH^- \rightleftharpoons HCO_2^- + H_2O$$
- (A)  $5.6 \times 10^{-11}$  (B)  $5.6 \times 10^3$   
(C)  $1.8 \times 10^{10}$  (D)  $5.6 \times 10^{17}$
14. Solid calcium carbonate is in equilibrium as described below:
- $$Ca^{2+}(aq) + 2HCO_3^-(aq) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(l)$$
- If  $CO_2(g)$  is added to this system in a rigid container, which changes will take place?
- I. Additional  $CaCO_3(s)$  will precipitate.  
II. The pH will increase.
- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II
15. Calcium carbonate occurs in two common crystalline forms, calcite ( $K_{sp} = 3.4 \times 10^{-9}$ ) and aragonite ( $K_{sp} = 6.0 \times 10^{-9}$ ). Which of these is most thermodynamically stable?
- (A) Calcite  
(B) Aragonite  
(C) The two forms are equally stable.  
(D) The relative stability cannot be determined from the  $K_{sp}$  values.
16. What is the pH of a 0.043 M solution of HI?
- (A) 0.04 (B) 1.37  
(C) 1.67 (D) 3.15
17. At 1200 K, the equilibrium constant for the reaction
- $$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$
- is  $K_p = 2.9 \times 10^{-7}$ . A mixture of 0.79 bar  $N_2$  and 0.21 bar  $O_2$  is prepared at 300 K and heated to 1200 K while maintaining a constant total pressure of 1.00 bar. What is the equilibrium partial pressure of  $NO(g)$  at 1200 K?
- (A)  $2.4 \times 10^{-8}$  bar (B)  $1.1 \times 10^{-4}$  bar  
(C)  $2.2 \times 10^{-4}$  bar (D)  $5.4 \times 10^{-4}$  bar
18. Sodium sulfate is slowly added to 1.00 L of a solution that is 0.10 M in both  $Pb^{2+}(aq)$  and  $Sr^{2+}(aq)$ . How much  $Na_2SO_4$  must be added to induce precipitation of  $SrSO_4$ ? The  $K_{sp}$  of  $PbSO_4$  is  $1.8 \times 10^{-8}$  and the  $K_{sp}$  of  $SrSO_4$  is  $3.5 \times 10^{-7}$ .
- (A)  $3.3 \times 10^{-6}$  mol (B)  $3.5 \times 10^{-6}$  mol  
(C)  $5.9 \times 10^{-4}$  mol (D)  $9.5 \times 10^{-2}$  mol
19. Lead(II) fluoride is sparingly soluble ( $K_{sp} = 4.0 \times 10^{-8}$ ). Which, when added to a saturated solution of  $PbF_2$ , would increase its solubility to the greatest extent?
- (A)  $KNO_3$  (B)  $Pb(NO_3)_2$   
(C)  $KF$  (D)  $HNO_3$
20. The pH of a 0.045 M solution of the monoprotic acid boric acid is 5.30. What is the  $K_a$  of boric acid?
- (A)  $1.1 \times 10^{-4}$  (B)  $5.0 \times 10^{-6}$   
(C)  $2.0 \times 10^{-9}$  (D)  $5.6 \times 10^{-10}$
21.  $NOBr$  decomposes to  $NO$  and  $Br_2$  gas in a closed container as shown.
- $$2 NOBr(g) \rightleftharpoons 2 NO(g) + Br_2(g) \quad \Delta H^\circ_{rxn} = +16.3 \text{ kJ mol}^{-1}$$
- Which changes will result in an increase in the number of moles of  $NO$  present at equilibrium?
- I. Increasing the temperature  
II. Decreasing the volume of the container
- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

# Equilibrium

22. At a certain temperature,  $\text{SO}_2\text{Cl}_2$  decomposes into  $\text{SO}_2$  and  $\text{Cl}_2$  as shown below.



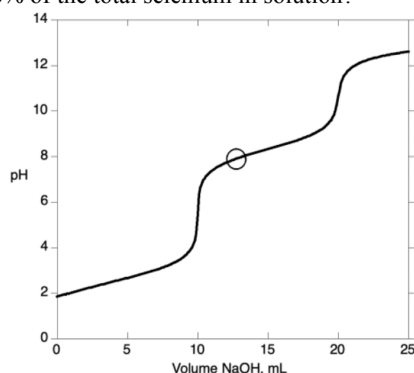
An equal number of moles of  $\text{SO}_2\text{Cl}_2$ ,  $\text{SO}_2$ , and Ar are introduced into a sealed flask to give a total pressure of 1.80 bar. The flask is maintained at the same temperature until equilibrium is attained. What is the final pressure of  $\text{Cl}_2(\text{g})$ ?

- (A) 0.23 bar (B) 0.27 bar  
(C) 0.35 bar (D) 0.50 bar

23. A buffer solution consists of 0.100 mol  $\text{Na}_3\text{PO}_4$  and 0.100 mol  $\text{Na}_2\text{HPO}_4$  dissolved in water. How much of which component of the buffer must be added to change the pH to 12.25? The  $\text{p}K_a$  of  $\text{HPO}_4^{2-}$  is 12.37.

- (A) 0.032 mol  $\text{Na}_2\text{HPO}_4$  (B) 0.032 mol  $\text{Na}_3\text{PO}_4$   
(C) 0.132 mol  $\text{Na}_2\text{HPO}_4$  (D) 0.132 mol  $\text{Na}_3\text{PO}_4$

24. A 0.100 M aqueous solution of  $\text{H}_2\text{SeO}_3$  is titrated with 1.000 M NaOH solution. At the point marked with a circle on the titration curve, which species comprise at least 10% of the total selenium in solution?



- (A)  $\text{H}_2\text{SeO}_3$  only (B)  $\text{H}_2\text{SeO}_3$  and  $\text{HSeO}_3^-$   
(C)  $\text{HSeO}_3^-$  only (D)  $\text{HSeO}_3^-$  and  $\text{SeO}_3^{2-}$

25. What mass of lead(II) fluoride ( $K_{sp} = 4.0 \times 10^{-8}$ ) will dissolve in 1.00 L of water?

- (A) 0.025 g (B) 0.049 g  
(C) 0.087 g (D) 0.53 g

26. The  $K_a$  of the pyridinium ion,  $\text{C}_5\text{H}_5\text{NH}^+$ , is  $5.9 \times 10^{-6}$ . What is the pH of a 0.170 M solution of pyridine,  $\text{C}_5\text{H}_5\text{N}$ ?

- (A) 3.00 (B) 4.77 (C) 8.77 (D) 9.23

27. Excess solid silver chloride is added to water and the mixture stirred until equilibrium is achieved. Addition of which substance will increase the concentration of chloride ion in this solution?

- (A)  $\text{AgNO}_3$  (B)  $\text{AgCl}$   
(C)  $\text{NH}_3$  (D)  $\text{H}_2\text{O}$

Problems 28 and 29 are about the exothermic reaction of iodine with iodide ion shown below.



28.  $1.0 \times 10^{-4}$  mol  $\text{I}_2$  and  $4.0 \times 10^{-3}$  mol KI are dissolved in water at  $25^\circ\text{C}$  to give 100.0 mL of solution. What is the equilibrium concentration of  $\text{I}_2$  in this solution?

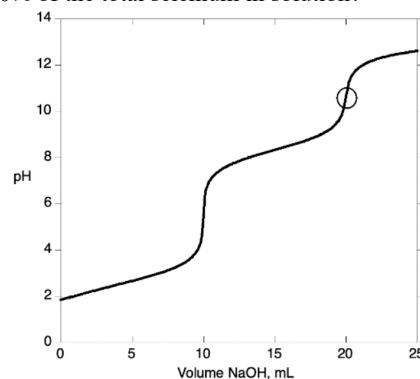
- (A)  $1.3 \times 10^{-6}$  M (B)  $3.4 \times 10^{-5}$  M  
(C)  $5.0 \times 10^{-4}$  M (D)  $5.3 \times 10^{-3}$  M

29. Which changes will result in an increase in the number of moles of  $\text{I}_2(\text{aq})$  present at equilibrium?

- I. Increasing the temperature  
II. Replacing the KI with an equal mass of NaI

- (A) I only (B) II only  
(C) Both I and II (D) Neither I nor II

30. A 0.100 M aqueous solution of  $\text{H}_2\text{SeO}_3$  is titrated with 1.000 M NaOH solution. At the point marked with a circle on the titration curve, which species comprise at least 10% of the total selenium in solution?



- (A)  $\text{H}_2\text{SeO}_3$  only (B)  $\text{H}_2\text{SeO}_3$  and  $\text{HSeO}_3^-$   
(C)  $\text{HSeO}_3^-$  and  $\text{SeO}_3^{2-}$  (D)  $\text{SeO}_3^{2-}$  only

31. Lead(II) iodide has a solubility product constant of  $1.4 \times 10^{-8}$ . What is the concentration of  $\text{I}^-$  in a saturated solution of  $\text{PbI}_2$ ?

- (A)  $1.2 \times 10^{-4}$  M (B)  $2.4 \times 10^{-4}$  M  
(C)  $1.5 \times 10^{-3}$  M (D)  $3.0 \times 10^{-3}$  M

# Equilibrium

32. Sodium bicarbonate decomposes at high temperatures according to the following equation:



At a certain temperature, 50.0 g of  $\text{NaHCO}_3$  is allowed to react in a 3.0 L evacuated vessel. At equilibrium, the total pressure is 6.25 atm, with some of both solids still present. If 100.0 g of  $\text{NaHCO}_3$  is allowed to react in the same vessel at the same temperature, what will the pressure be at equilibrium?

- (A) 6.25 atm                      (B) 8.84 atm  
(C) 12.5 atm                      (D) 25.0 atm

33. In a solution prepared by dissolving 0.1 mol  $\text{NaNO}_2$  in 1.0 L pure water, which species has the lowest concentration?

- (A)  $\text{Na}^+$       (B)  $\text{NO}_2^-$       (C)  $\text{HNO}_2$       (D)  $\text{H}_3\text{O}^+$

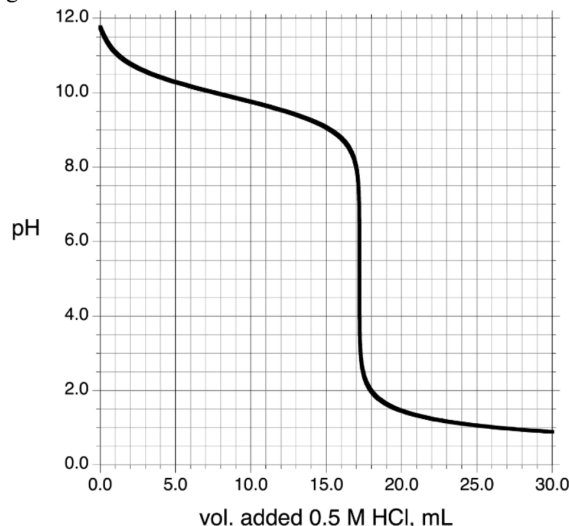
34. A saturated solution of which silver salt has the highest concentration of  $\text{Ag}^+$ ?

- (A)  $\text{AgCl}$ ,  $K_{\text{sp}} = 1.8 \times 10^{-10}$   
(B)  $\text{Ag}_2\text{CrO}_4$ ,  $K_{\text{sp}} = 1.1 \times 10^{-12}$   
(C)  $\text{AgBr}$ ,  $K_{\text{sp}} = 5.0 \times 10^{-13}$   
(D)  $\text{Ag}_2\text{SO}_3$ ,  $K_{\text{sp}} = 1.5 \times 10^{-14}$

35. What is the pH of a 0.10 M solution of  $\text{NH}_4\text{F}$ ? The  $K_a$  of  $\text{NH}_4^+$  is  $5.6 \times 10^{-10}$  and the  $K_a$  of  $\text{HF}$  is  $6.8 \times 10^{-4}$ .

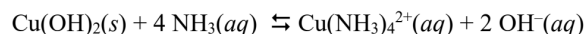
- (A) 2.08      (B) 5.12      (C) 6.21      (D) 8.08

36. 1.000 g of which salt, when titrated with 0.500 M  $\text{HCl}$ , gives the titration curve shown?



- (A) Potassium cyanide,  $\text{KCN}$  ( $M = 65.12$ ,  $\text{p}K_a$  of  $\text{HCN} = 9.2$ )  
(B) Sodium nitrite,  $\text{NaNO}_2$  ( $M = 69.00$ ,  $\text{p}K_a$  of  $\text{HNO}_2 = 3.4$ )  
(C) Potassium phenoxide,  $\text{K}(\text{C}_6\text{H}_5\text{O})$  ( $M = 116.20$ ,  $\text{p}K_a$  of  $\text{C}_6\text{H}_5\text{OH} = 9.9$ )  
(D) Sodium pivalate,  $\text{Na}(\text{C}_4\text{H}_9\text{COO})$  ( $M = 124.11$ ,  $\text{p}K_a$  of  $\text{C}_4\text{H}_9\text{COOH} = 4.9$ )

37. What is the correct equilibrium expression for the given reaction?



- (A)  $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][\text{OH}^-]}{[\text{NH}_3]}$   
(B)  $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][2 \text{OH}^-]}{[4 \text{NH}_3]}$   
(C)  $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][\text{OH}^-]^2}{[\text{NH}_3]^4}$   
(D)  $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][2 \text{OH}^-]^2}{[4 \text{NH}_3]^4}$

38. 0.100 mol of  $\text{HF}$  ( $K_a = 6.6 \times 10^{-4}$ ) is added to water to make 1.00 L of solution. Which statement is correct at equilibrium?

- (A)  $[\text{H}_3\text{O}^+] = [\text{HF}]$                       (B)  $[\text{H}_3\text{O}^+] = 0.100 \text{ M}$   
(C)  $[\text{HF}] > [\text{F}^-]$                       (D)  $[\text{H}_3\text{O}^+] > [\text{HF}]$

# Equilibrium

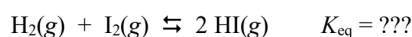
39. What is the solubility of silver oxalate,  $\text{Ag}_2\text{C}_2\text{O}_4$  ( $K_{\text{sp}} = 3.5 \times 10^{-11}$ ), in a 0.050 M sodium oxalate solution?

(A)  $1.4 \times 10^{-8}$  M      (B)  $5.9 \times 10^{-6}$  M  
(C)  $1.3 \times 10^{-5}$  M      (D)  $2.1 \times 10^{-4}$  M

40. A pure sample of a monoprotic acid is dissolved in water. The sample is titrated with sodium hydroxide solution. At the point where 20.0 mL of the NaOH solution has been added, the pH is 4.15. The phenolphthalein endpoint of the titration is observed when 50.0 mL of NaOH have been added. What is the  $\text{p}K_{\text{a}}$  of the acid?

(A) 4.15      (B) 4.33      (C) 4.55      (D) 5.19

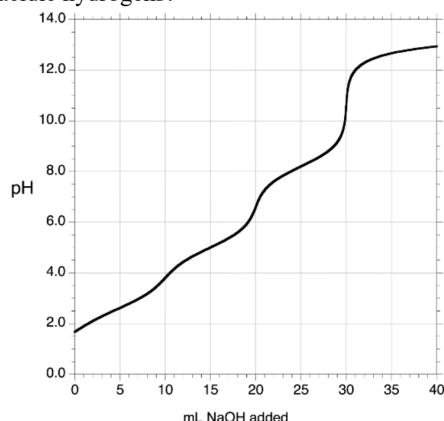
41. Consider the reaction:



Into a 1.00 L vessel, 1.00 mol  $\text{H}_2(\text{g})$  and 1.00 mol  $\text{I}_2(\text{g})$  are placed at a high temperature. When the reaction mixture stops changing, it is found that 79.0% of the  $\text{H}_2(\text{g})$  has reacted. What is the equilibrium constant for this reaction at this temperature?

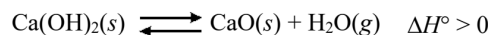
(A) 14.2      (B) 17.9      (C) 35.8      (D) 56.6

42. A triprotic acid is titrated with sodium hydroxide to give the titration curve below. What are the  $\text{p}K_{\text{a}}$  values of the three acidic hydrogens?



(A) 2.6, 5.0, 8.2      (B) 3.8, 6.6, 10.6  
(C) 3.8, 8.2, 12.7      (D) 5.0, 8.2, 12.7

43. The endothermic reaction shown below is at equilibrium in a sealed flask, with significant amounts of both  $\text{Ca}(\text{OH})_2(\text{s})$  and  $\text{CaO}(\text{s})$  present. Which action will increase the amount of  $\text{Ca}(\text{OH})_2(\text{s})$  at equilibrium?

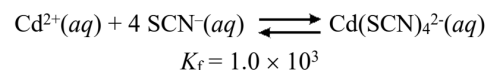


(A) Crushing the  $\text{Ca}(\text{OH})_2(\text{s})$  into smaller pieces  
(B) Decreasing the temperature of the flask  
(C) Adding more  $\text{CaO}(\text{s})$  to the flask  
(D) Adding  $\text{N}_2(\text{g})$  to the flask

44. A solution of ammonia,  $\text{NH}_3$ , has pH = 11.50. What is the ammonia concentration? (The  $\text{p}K_{\text{a}}$  of  $\text{NH}_4^+$  is 9.24.)

(A)  $1.7 \times 10^{-5}$  M      (B)  $3.2 \times 10^{-3}$  M  
(C)  $5.5 \times 10^{-3}$  M      (D) 0.58 M

45. At equilibrium, a solution contains 0.50 mol each of  $\text{Cd}^{2+}(\text{aq})$ ,  $\text{SCN}^-(\text{aq})$ , and  $\text{Cd}(\text{SCN})_4^{2-}(\text{aq})$ . What is the solution's volume?

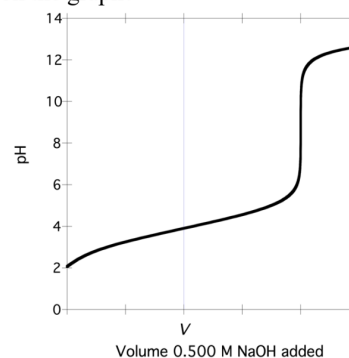


(A) 2.0 L      (B) 2.2 L      (C) 2.8 L      (D) 8.0 L

46. Barium fluoride has  $K_{\text{sp}} = 1.8 \times 10^{-7}$ . What is the maximum fluoride ion concentration possible in a solution with  $[\text{Ba}^{2+}] = 5.0 \times 10^{-4}$  M?

(A)  $3.6 \times 10^{-4}$  M      (B)  $3.6 \times 10^{-3}$  M  
(C)  $9.5 \times 10^{-3}$  M      (D)  $1.9 \times 10^{-2}$  M

47. Trimesic acid is a triprotic acid with  $\text{p}K_1 = 3.1$ ,  $\text{p}K_2 = 3.9$ , and  $\text{p}K_3 = 4.7$ . 100.0 mL of 0.100 M trimesic acid is titrated with 0.500 M  $\text{NaOH}(\text{aq})$  to give the graph below. What volume of  $\text{NaOH}(\text{aq})$  has been added at the point marked V on the graph?



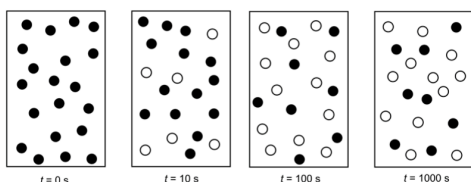
(A) 5 mL      (B) 10 mL      (C) 20 mL      (D) 30 mL

# Equilibrium

48. A solution containing  $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$ ,  $\text{CoCl}_4^{2-}(\text{aq})$ , and  $\text{Cl}^-(\text{aq})$  at equilibrium at room temperature is initially pink. When heated, the solution turns blue. Then, when  $\text{Ag}^+(\text{aq})$  is added, the solution turns back to pink. Which statements are correct?

I.  $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$  is pink.  
 II. Formation of  $\text{CoCl}_4^{2-}(\text{aq})$  from  $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$  and  $\text{Cl}^-(\text{aq})$  is exothermic.

- (A) I only (B) II only  
 (C) Both I and II (D) Neither I nor II
49. The diagrams below represent the progress of a reaction of  $\text{A} \rightarrow \text{B}$ , with molecules of A represented by solid circles and molecules of B represented by open circles. Which statement best describes the reaction at  $t = 1000$  s?



- (A) The reaction is at equilibrium because the concentrations of A and B are no longer changing.  
 (B) The reaction is not at equilibrium because not all of the A has been consumed.  
 (C) The reaction is not at equilibrium because the concentrations of A and B are not equal.  
 (D) The reaction is not at equilibrium because the positions of the molecules are still changing.
50. What is the pH of a 0.10 M solution of NaCN? The  $K_a$  of HCN is  $4.9 \times 10^{-10}$ .  
 (A) 2.85 (B) 5.15 (C) 8.85 (D) 11.15

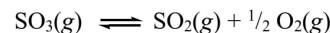
51. For which of the following reactions would the yield of products at equilibrium NOT increase at a higher pressure?

(A)  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g})$   
 (B)  $\text{Ti}(\text{s}) + 2 \text{Cl}_2(\text{g}) \rightleftharpoons \text{TiCl}_4(\text{g})$   
 (C)  $2 \text{C}_2\text{H}_4(\text{g}) + 2 \text{H}_2\text{O}(\text{g}) \rightleftharpoons 2 \text{C}_2\text{H}_6(\text{g}) + \text{O}_2(\text{g})$   
 (D)  $4 \text{HCl}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{H}_2\text{O}(\text{l}) + 2 \text{Cl}_2(\text{g})$

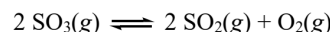
52. Chromium(III) hydroxide has  $K_{sp} = 1.6 \times 10^{-30}$ . What is the molar solubility of  $\text{Cr}(\text{OH})_3$  in a solution whose pH is maintained at 6.00?

(A)  $1.6 \times 10^{-12} \text{ mol L}^{-1}$  (B)  $1.6 \times 10^{-8} \text{ mol L}^{-1}$   
 (C)  $3.6 \times 10^{-8} \text{ mol L}^{-1}$  (D)  $1.6 \times 10^{-6} \text{ mol L}^{-1}$

53. At 400 K, this reaction has  $K_p = 8.2 \times 10^{-4}$ .

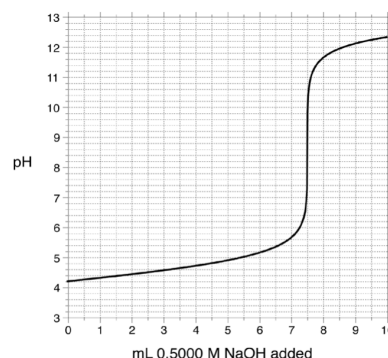


What is  $K_p$  at 400 K for the following reaction?



- (A)  $6.7 \times 10^{-7}$  (B)  $8.2 \times 10^{-4}$   
 (C)  $1.6 \times 10^{-3}$  (D)  $2.9 \times 10^{-2}$

54. An unknown monoprotic carboxylic acid RCOOH is mixed in a 1:1 mole ratio with its sodium salt, Na(RCOO). A 1.000 g sample of this 1:1 mixture is dissolved in 50 mL water and titrated with 0.5000 M NaOH solution. The pH is measured as a function of added titrant to give the following titration curve:



Which statements are correct?

- I. The molar mass of RCOOH is within 5% of  $265 \text{ g mol}^{-1}$ .  
 II. The  $\text{p}K_a$  of RCOOH is within 0.1 unit of 4.2.

- (A) I only (B) II only  
 (C) Both I and II (D) Neither I nor II

55. What mass of silver chloride ( $M = 143.4$ ) will dissolve in 1.00 L of water? The  $K_{sp}$  of AgCl is  $1.8 \times 10^{-10}$ .

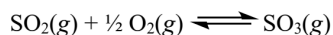
(A) 1.4 mg (B) 1.9 mg (C) 2.9 mg (D) 3.8 mg

56. What is the pH of a 0.20 M solution of sodium benzoate,  $\text{Na}(\text{C}_6\text{H}_5\text{COO})$ ? The  $K_a$  of benzoic acid,  $\text{C}_6\text{H}_5\text{COOH}$ , is  $6.5 \times 10^{-5}$ .

(A) 5.26 (B) 8.74 (C) 9.09 (D) 11.56

# Equilibrium

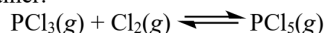
57. Sulfur trioxide is formed from the reaction of sulfur dioxide and oxygen:



At 1000 K, an equilibrium mixture has partial pressures of 0.562 bar  $\text{SO}_2$ , 0.101 bar  $\text{O}_2$ , and 0.332 bar  $\text{SO}_3$ . What is the equilibrium constant  $K_p$  for the reaction at this temperature?

- (A) 1.86      (B) 3.46      (C) 5.85      (D) 16.8

58. The following endothermic reaction is at equilibrium in a sealed container.



Which of the following changes would result in an increase in the number of moles of  $\text{PCl}_5(\text{g})$  present at equilibrium?

- I. Increasing the temperature    II. Increasing the volume

- (A) I only                      (B) II only  
(C) Both I and II            (D) Neither I nor II

59. A 0.12 M solution of a monoprotic acid is 2.3% ionized. What is the  $K_a$  of this acid?

- (A)  $2.8 \times 10^{-3}$                       (B)  $3.3 \times 10^{-4}$   
(C)  $6.5 \times 10^{-5}$                       (D)  $7.6 \times 10^{-6}$

60. Calcium fluoride,  $\text{CaF}_2$ , has a molar solubility of  $2.1 \times 10^{-4} \text{ mol L}^{-1}$  at pH = 7.00. By what factor does its molar solubility increase in a solution with pH = 3.00? The  $pK_a$  of HF is 3.17.

- (A) 1.48      (B) 1.83      (C) 2.48      (D) 4.96

61. Which solution has the greatest percent ionization?

- (A) 0.010 M formic acid ( $K_a = 1.8 \times 10^{-4}$ )  
(B) 0.10 M formic acid ( $K_a = 1.8 \times 10^{-4}$ )  
(C) 0.010 M acetic acid ( $K_a = 1.8 \times 10^{-5}$ )  
(D) 0.10 M acetic acid ( $K_a = 1.8 \times 10^{-5}$ )

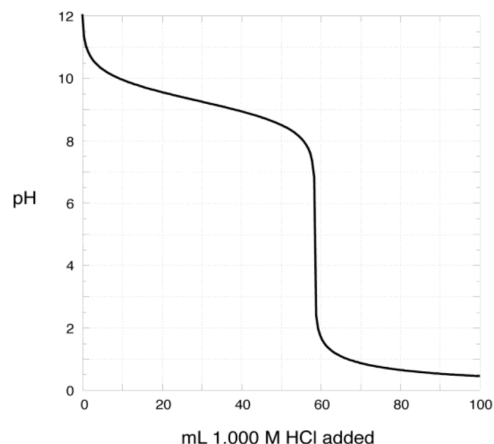
62. Calcium oxalate,  $\text{CaC}_2\text{O}_4$  ( $M = 128.1$ ), dissolves to the extent of  $0.67 \text{ mg L}^{-1}$ . What is its  $K_{sp}$ ?

- (A)  $6.7 \times 10^{-4}$                       (B)  $4.5 \times 10^{-7}$   
(C)  $2.7 \times 10^{-11}$                       (D)  $5.7 \times 10^{-16}$

63. What is the concentration of a solution of  $\text{K}_2\text{CO}_3$  that has pH = 11.90? (For  $\text{H}_2\text{CO}_3$ ,  $K_{a1} = 4.2 \times 10^{-7}$ ,  $K_{a2} = 4.8 \times 10^{-11}$ .)

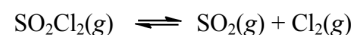
- (A)  $3.0 \times 10^{-1} \text{ M}$                       (B)  $2.6 \times 10^{-2} \text{ M}$   
(C)  $7.9 \times 10^{-3} \text{ M}$                       (D)  $1.3 \times 10^{-12} \text{ M}$

64. 1.000 g of a weak base is titrated with 1.000 M aqueous HCl to give the data shown. What is the identity of the base?



- (A) Ammonia,  $\text{NH}_3$  ( $pK_a$  of  $\text{NH}_4^+ = 9.3$ )  
(B) Aniline,  $\text{C}_6\text{H}_5\text{NH}_2$  ( $pK_a$  of  $\text{C}_6\text{H}_5\text{NH}_3^+ = 4.6$ )  
(C) Hydroxylamine,  $\text{NH}_2\text{OH}$  ( $pK_a$  of  $\text{NH}_3\text{OH}^+ = 6.0$ )  
(D) Trimethylamine,  $(\text{CH}_3)_3\text{N}$  ( $pK_a$  of  $(\text{CH}_3)_3\text{NH}^+ = 9.8$ )

65. Sulfuryl chloride is in equilibrium with sulfur dioxide and chlorine gas:



A system with a volume of 1.00 L is in equilibrium at a certain temperature with  $p(\text{SO}_2\text{Cl}_2) = 1.00 \text{ bar}$  and  $p(\text{SO}_2) = p(\text{Cl}_2) = 0.10 \text{ bar}$ . By how much will the number of moles of  $\text{SO}_2\text{Cl}_2$  at equilibrium change if the volume is reduced to 0.50 L?

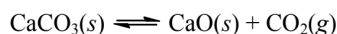
- (A) Increase 1-10%                      (B) Increase 11-50%  
(C) Decrease 1-10%                      (D) Decrease 11-50%

66. Which of the following will increase  $[\text{Cl}^-]$  in a saturated solution of  $\text{AgCl}$  in contact with excess solid silver chloride?

- I. Addition of  $\text{AgCl}(s)$                       II. Addition of  $\text{NH}_3(aq)$   
(A) I only                                      (B) II only  
(C) Both I and II                              (D) Neither I nor II

# Equilibrium

67. A 60.0 g sample of  $\text{CaCO}_3$  is heated to 950 K in a 1.00 L evacuated container, where it reacts according to the following equation:



After equilibrium is attained, the pressure of  $\text{CO}_2(g)$  is 30.0 mm Hg. When the experiment is repeated using 120.0 g  $\text{CaCO}_3$ , what is the equilibrium pressure  $P$ ?

- (A) 15.0 mm Hg  $\leq P < 30.0$  mm Hg  
(B)  $P = 30.0$  mm Hg  
(C) 30.0 mm Hg  $< P < 60.0$  mm Hg  
(D)  $P = 60.0$  mm Hg

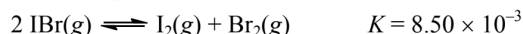
68. Nitrous acid,  $\text{HNO}_2$ , has  $K_a = 4.5 \times 10^{-4}$ . What is the best description of the species present in a 0.1 M solution of nitrous acid?

- (A)  $\text{HNO}_2(aq)$  is the predominant species; much smaller amounts of  $\text{H}^+(aq)$  and  $\text{NO}_2^-(aq)$  exist.  
(B)  $\text{H}^+(aq)$  and  $\text{NO}_2^-(aq)$  are the predominant species; much smaller amounts of  $\text{HNO}_2(aq)$  exist.  
(C) Only  $\text{H}^+(aq)$  and  $\text{NO}_2^-(aq)$  are present in measurable amounts.  
(D)  $\text{HNO}_2(aq)$ ,  $\text{H}^+(aq)$ , and  $\text{NO}_2^-(aq)$  are all present in comparable amounts.

69. Silver sulfate,  $\text{Ag}_2\text{SO}_4$  ( $M = 311.87$ ), has  $K_{sp} = 6.0 \times 10^{-5}$ . What mass of  $\text{Ag}_2\text{SO}_4$  will dissolve per liter of water?

- (A) 0.019 g                      (B) 1.7 g  
(C) 2.4 g                      (D) 7.7 g

70.  $\text{IBr}(g)$  is in equilibrium with  $\text{I}_2(g)$  and  $\text{Br}_2(g)$  at 150 °C:



Initially, a closed vessel at 150 °C has a partial pressure of  $\text{IBr}$  of 0.350 atm and partial pressures of  $\text{I}_2$  and  $\text{Br}_2$  each of 0.750 atm. What is the partial pressure of  $\text{IBr}$  once the system reaches equilibrium?

- (A) 1.01 atm                      (B) 1.09 atm  
(C) 1.56 atm                      (D) 1.82 atm

71. 20 mL of an approximately 10% aqueous solution of ethylamine,  $\text{CH}_3\text{CH}_2\text{NH}_2$ , is titrated with 0.3000 M aqueous  $\text{HCl}$ . Which indicator would be most suitable for this titration? The  $\text{p}K_a$  of  $\text{CH}_3\text{CH}_2\text{NH}_3^+$  is 10.75.

- (A) Thymol blue, color change from pH = 1.2 to 2.8  
(B) Bromocresol green, color change from pH = 4.0 to 5.6  
(C) Phenolphthalein, color change from pH = 8.0 to 10.0  
(D) Alizarin yellow R, color change from pH = 10.0 to 12.0

72. The tetramminecopper(II) ion,  $\text{Cu}(\text{NH}_3)_4^{2+}$ , has a formation constant  $K_f = 1.1 \times 10^{13}$ . What is the minimum concentration of free ammonia in solution required to ensure that at least 99.9% of the dissolved copper(II) ion is found in the form of its ammonia complex?

- (A)  $9 \times 10^{-14}$  M                      (B)  $9 \times 10^{-11}$  M  
(C)  $8 \times 10^{-4}$  M                      (D)  $3 \times 10^{-3}$  M